Biblatex to Bibtex for Sphinx

Rohit Goswami · 2021-05-23 · tools · workflow · tex · documentation

Background

> *Transitioning from `biblatex` to `bibtex` with `biber` for `sphinx`*

I recently started a set of notes using `jupyter-book`. However, in the process I ran into a horrific bibliography related SNAFU. `sphinx` in its infinite wisdom only accepts a rather odd subset of `bibtex`.

I have been happily exporting my giant bibliography with Zotero (and better bibtex, synced via WebDAV) exporting my references as `biblatex` while `sphinx` started choking dreadfully. This post describes attempts to reconcile the `biblatex` sources without manual intervention.

Input

The input file was an innocuous looking tiny neat `biblatex` file:

```
@online{grasedyckParameterdependentSmootherMultigrid2020,
  title = {A Parameter-Dependent Smoother for the Multigrid Method},
  author = {Grasedyck, Lars and Klever, Maren and Löbbert, Christian and Werthmann, Tim A.},
  date = {2020-08-03},
  url = {http://arxiv.org/abs/2008.00927},
  urldate = {2021-05-22},
  archiveprefix = {arXiv},
  eprint = {2008.00927},
  eprinttype = {arxiv},
  keywords = {65N55; 15A69,Mathematics - Numerical Analysis},
  primaryclass = {cs, math}
}

@article{grasedyckDistributedHierarchicalSVD2018,
  title = {Distributed Hierarchical {{SVD}} in the {{Hierarchical Tucker}} Format},
  author = {Grasedyck, Lars and Löbbert, Christian},
  date = {2018},
  journaltitle = {Numerical Linear Algebra with Applications},
  volume = {25},
  pages = {e2174},
  issn = {1099-1506},
  doi = {10.1002/nla.2174},
  url = {https://onlinelibrary.wiley.com/doi/abs/10.1002/nla.2174},
  urldate = {2021-05-22},
  annotation = {\_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/nla.2174},
  keywords = {Hierarchical Tucker,HT,multigrid,parallel algorithms,SVD,tensor arithmetic},
  langid = {english},
  number = {6}
}

@article{grasedyckHierarchicalSingularValue2010,
  title = {Hierarchical {{Singular Value Decomposition}} of {{Tensors}}},
  author = {Grasedyck, Lars},
  date = {2010-01-01},
  journaltitle = {SIAM Journal on Matrix Analysis and Applications},
  shortjournal = {SIAM J. Matrix Anal. Appl.},
  volume = {31},
  pages = {2029--2054},
  publisher = {{Society for Industrial and Applied Mathematics}},
  issn = {0895-4798},
  doi = {10.1137/090764189},
  url = {https://epubs.siam.org/doi/abs/10.1137/090764189},
  urldate = {2021-05-22},
  number = {4}
}
```

Manual cleaning

The first attempt was with `bibutils`.

```
biblatex2xml references.bib | xml2bib | tee  refs.bib
```

`sphinx` does not support `electronic` and demands `misc` instead.

```
biblatex2xml references.bib | xml2bib | sed -e "s/Electronic{/misc{/g" | tee  refs.bib
```

This brought us to:

```
@misc{grasedyckParameterdependentSmootherMultigrid2020,
author="Grasedyck, Lars
and Klever, Maren
and L{\"o}bbert, Christian
and Werthmann, Tim A.",
title="A Parameter-Dependent Smoother for the Multigrid Method",
year="2020",
month="Aug",
day="03",
archivePrefix="arXiv",
eprint="2008.00927",
url="http://arxiv.org/abs/2008.00927"
}

@Article{grasedyckDistributedHierarchicalSVD2018,
author="Grasedyck, Lars
and L{\"o}bbert, Christian",
title="Distributed Hierarchical SVD in the Hierarchical Tucker Format",
journal="Numerical Linear Algebra with Applications",
year="2018",
volume="25",
number="6",
pages="e2174",
issn="1099-1506",
doi="10.1002/nla.2174",
url="https://onlinelibrary.wiley.com/doi/abs/10.1002/nla.2174",
url="https://doi.org/10.1002/nla.2174"
}

@Article{grasedyckHierarchicalSingularValue2010,
author="Grasedyck, Lars",
title="Hierarchical Singular Value Decomposition of Tensors",
journal="SIAM Journal on Matrix Analysis and Applications",
year="2010",
month="Jan",
day="01",
volume="31",
number="4",
pages="2029--2054",
issn="0895-4798",
doi="10.1137/090764189",
url="https://epubs.siam.org/doi/abs/10.1137/090764189",
url="https://doi.org/10.1137/090764189"
}
```

Unfortunately this also generated multiple `url` lines, which choke `sphinx` of course. Additionally, the workflow is rather fickle e.g. not starting the file with a blank line causes the first reference to be silently skipped.

Biber

The concept here is that `biber` can take a configuration file to process the `bib` file and is also able to copy the formatted file over with `tool`. So our solution will boil down to writing a configuration file and then using it.

First we get the location of the `configuration` (also check the `version`).

```
biber --tool-config
biber --version
```

Then we inspect it and extract relevant entries to a configuration file of our own, with some modifications.

- We need to reverse each entry as described here
    - However in practice we can get away with a minimal mapping and some command line options..
- We would also like to be mindful of the mapping from `online` to `misc`

```
<?xml version="1.0" encoding= "utf-8"?>
<config>
  <sourcemap>
    <maps datatype="bibtex">
      <!-- Easy type conversions -->
      <map>
        <map_step map_type_source="report" map_type_target="techreport"/>
        <map_step map_type_source="online" map_type_target="misc"/>
      </map>
    </maps>
  </sourcemap>
</config>
```

At this stage, we are partially done, but the resulting file is rather ugly, and **most importantly** dates have not yet been transformed. This SE answer provided a hint, as did this gist. Effectively, we need to overwrite part of the map.

```
<!-- Date to year, month -->
<map>
  <map_step map_field_source="date"
  map_field_target="year" />
</map>
<map>
  <map_step map_field_source="year"
  map_match="(\d{4}|\d{2})-(\d{1,2})-(\d{1,2})"
  map_final="1" />
  <map_step map_field_source="year"
  map_match="(\d{4}|\d{2})-(\d{1,2})-(\d{1,2})"
  map_replace="$1" />
  <map_step map_field_set="month" map_origfieldval="1" />
  <map_step map_field_source="month"
  map_match="(\d{4}|\d{2})-(\d{1,2})-(\d{1,2})"
  map_replace="$2" />
</map>
<map>
  <map_step map_field_source="year"
  map_match="(\d{4}|\d{2})-(\d{1,2})" map_final="1" />
  <map_step map_field_source="year"
  map_match="(\d{4}|\d{2})-(\d{1,2})" map_replace="$1" />
  <map_step map_field_set="month" map_origfieldval="1" />
  <map_step map_field_source="month"
  map_match="(\d{4}|\d{2})-(\d{1,2})" map_replace="$2" />
</map>
```

Now we can also add some pretty printing (before the `sourcemap`):

```
<output_fieldcase>lower</output_fieldcase>
<output_resolve>1</output_resolve>
<output_safechars>1</output_safechars>
<output_format>bibtex</output_format>
```

Putting it all together:

```
<?xml version="1.0" encoding="utf-8"?>
<!-- Got the date from
https://gist.githubusercontent.com/mkouhia/f00fea7fc8d4effd9dfd/raw/500e9dbc6aa43a47e39c45ba230738ff4544709f/b
iblatex-to-bibtex.conf -->
<config>
  <output_fieldcase>lower</output_fieldcase>
  <output_resolve>1</output_resolve>
  <output_safechars>1</output_safechars>
  <output_format>bibtex</output_format>
  <sourcemap>
    <maps datatype="bibtex">
      <!-- Easy type conversions -->
      <map>
        <map_step map_type_source="report" map_type_target="techreport"/>
        <map_step map_type_source="online" map_type_target="misc"/>
      </map>
      <!-- Date to year, month -->
      <map>
        <map_step map_field_source="date"
        map_field_target="year" />
      </map>
      <map>
        <map_step map_field_source="year"
        map_match="(\d{4}|\d{2})-(\d{1,2})-(\d{1,2})"
        map_final="1" />
        <map_step map_field_source="year"
        map_match="(\d{4}|\d{2})-(\d{1,2})-(\d{1,2})"
        map_replace="$1" />
        <map_step map_field_set="month" map_origfieldval="1" />
        <map_step map_field_source="month"
        map_match="(\d{4}|\d{2})-(\d{1,2})-(\d{1,2})"
        map_replace="$2" />
      </map>
      <map>
        <map_step map_field_source="year"
        map_match="(\d{4}|\d{2})-(\d{1,2})" map_final="1" />
        <map_step map_field_source="year"
        map_match="(\d{4}|\d{2})-(\d{1,2})" map_replace="$1" />
        <map_step map_field_set="month" map_origfieldval="1" />
        <map_step map_field_source="month"
        map_match="(\d{4}|\d{2})-(\d{1,2})" map_replace="$2" />
      </map>
    </maps>
  </sourcemap>
</config>
```

However, we have not yet mapped `journaltitle` and `location`. These are now easier to set on the command line so we shall do so. Also note that in order to have the `date` logic work, we will need the `--output-legacy-date` option as well.

```
# Runs
biber --tool --configfile=biberConf.xml references.bib --output-file refsTmp.bib --output-legacy-date --
output-field-replace=location:address,journaltitle:journal
```

This gives a `sphinx` compatible file:

```
@misc{grasedyckParameterdependentSmootherMultigrid2020,
  author = {Grasedyck, Lars and Klever, Maren and L\"{o}bbert, Christian and Werthmann, Tim A.},
  url = {http://arxiv.org/abs/2008.00927},
  eprint = {2008.00927},
  eprinttype = {arxiv},
  keywords = {65N55; 15A69,Mathematics - Numerical Analysis},
  month = {08},
  title = {A Parameter-Dependent Smoother for the Multigrid Method},
  urldate = {2021-05-22},
  year = {2020},
}

@article{grasedyckDistributedHierarchicalSVD2018,
  author = {Grasedyck, Lars and L\"{o}bbert, Christian},
  url = {https://onlinelibrary.wiley.com/doi/abs/10.1002/nla.2174},
  annotation = {\_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/nla.2174},
  doi = {10.1002/nla.2174},
  issn = {1099-1506},
  journal = {Numerical Linear Algebra with Applications},
  keywords = {Hierarchical Tucker,HT,multigrid,parallel algorithms,SVD,tensor arithmetic},
  langid = {english},
  number = {6},
  pages = {e2174},
  title = {Distributed Hierarchical {{SVD}} in the {{Hierarchical Tucker}} Format},
  urldate = {2021-05-22},
  volume = {25},
  year = {2018},
}

@article{grasedyckHierarchicalSingularValue2010,
  author = {Grasedyck, Lars},
  publisher = {Society for Industrial and Applied Mathematics},
  url = {https://epubs.siam.org/doi/abs/10.1137/090764189},
  doi = {10.1137/090764189},
  issn = {0895-4798},
  journal = {SIAM Journal on Matrix Analysis and Applications},
  month = {01},
  number = {4},
  pages = {2029--2054},
  shortjournal = {SIAM J. Matrix Anal. Appl.},
  title = {Hierarchical {{Singular Value Decomposition}} of {{Tensors}}},
  urldate = {2021-05-22},
  volume = {31},
  year = {2010},
}
```

Conclusions

This was a rather annoying excursion into the unpleasant underbelly of `bibtex` and `biblatex`. The `biber` solution is actually rather elegant, but most definitely not immediately obvious.

Reading list

Bibliography tooling and the citation formats this conversion moves between, collected as a citation list: 3 references. The collection exports to BibTeX or any CSL style, so it can go straight into a bibliography.